

Nikhil Mourya

Jaipur, India

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ML engineer (graduating May 2027) with 6 months of production experience in multi-agent orchestration, RAG systems, LLM fine-tuning, and MLOps deployment on AWS. Shipped systems processing **10K+ requests/day**; proficient in PyTorch, LangGraph, Hugging Face Transformers, LoRA/PEFT, and Docker. IIT Kharagpur research collaboration. Codeforces Specialist (1400+).

Experience

HireBuddy

May 2025 – Nov 2025

Software Engineer – Machine Learning

Hybrid

- Built an India-specific resume scoring system using **RAG pipelines**, transformer embeddings, and **FAISS** vector search, fine-tuned on a **75K-resume** Kaggle dataset; achieved **35% improvement in shortlist precision** vs. keyword-matching baseline at **10K+ resumes/day** throughput.
- Designed end-to-end RAG retrieval pipeline for scoring, feedback generation, and candidate retrieval using **LangChain**, **LangGraph**, and **OpenAI API**; implemented evaluation using **RAGAS**, **LangSmith**, and ROUGE-based metrics to measure retrieval quality, ranking consistency, and response grounding.
- Deployed **MLOps** pipeline to production via **Docker** and **AWS**, serving **10K+ daily requests** through a REST API; model serving integration into recruiter workflows increased platform engagement by **40%**.

Projects

HiveMind – Autonomous Deep Research System

May 2026

- Built a production-grade autonomous research system with a **5-agent sequential pipeline** (Planner → Researcher → Critic → Writer → Evaluator); orchestrator decomposes queries into tasks, retrieves evidence via **ChromaDB + cosine reranking**, critiques findings, and autonomously loops until a **deterministic confidence threshold** (mean retrieval score) is met – preventing LLM hallucinated confidence on weak evidence.
- Enforced single-threaded LLM execution via **asyncio.Lock** to prevent OOM on local hardware and Groq rate-limit bursts; containerized with **Docker Compose** (FastAPI + **Redis** task queue + ChromaDB + **OpenTelemetry** distributed tracing); SSE streaming for real-time pipeline state updates.

CodeLens – Offline Semantic Code Search (VS Code Extension)

Apr 2026

- Built a fully offline VS Code extension for natural-language code search; indexes **9K+ chunks** via **AST-based chunking** (tree-sitter) and 768-dim sentence embeddings – no cloud, no API keys; **sub-10ms** ANN query latency via FastAPI backend.
- Auto-reindexes on file save using MD5 content hashing; semantic search accuracy consistently outperformed keyword-based grep on a 50-query internal benchmark.

PEGASUS + LoRA: Efficient News Summarization (XSum)

Feb 2026

- Fine-tuned PEGASUS via **LoRA/PEFT** on BBC XSum: reduced trainable parameters **767M → 1.57M (99.8%)** with **27x faster training** (260s → 9.5s); full fine-tuning produced incoherent outputs on unseen domains – LoRA generalized correctly, validating **PEFT for low-resource summarization**.

Pruned U-Net for Biomedical Image Segmentation

Mar 2025

- Research collaboration with IIT Kharagpur. **Structured pruning: 97.3% parameter reduction, 92% FLOPs reduction, IoU > 0.95** on MoNuSeg – compression held until ~96% reduction then degraded sharply, establishing a practical compression ceiling.

Hackathons

Smart India Hackathon – Finalist

Oct 2024

ML & Backend

Delhi, India

- Hospital queue prioritization model; **28% reduction** in estimated patient wait time. Built explainability layer so predictions were interpretable to non-technical clinical staff – accuracy alone was not the acceptance criterion.

Technical Skills

Languages: Python, C++, SQL

ML & LLMs: PyTorch, TensorFlow, Hugging Face Transformers, LoRA/PEFT, LangGraph, LangChain, RAG pipelines, LLM fine-tuning, prompt engineering, model compression

LLM Infra & Eval: OpenAI API, Groq API, Ollama, RAGAS, LangSmith, precision/recall

Infra & MLOps: FastAPI, Docker, AWS, Git, FAISS, ChromaDB, Redis, tree-sitter, REST APIs, model deployment/serving, asyncio

Observability: OpenTelemetry, Pydantic

Competitive Programming: Codeforces Specialist (1400+)

Education

Birla Institute of Technology, Mesra

Aug 2023 – May 2027

B.Tech in Physics and Computing | Focus: ML, Deep Learning, Optimization

Jaipur, Rajasthan